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Instructions:

- **Print the question paper** and start writing your answers as instructed.
- **Do not print the questions on both sides of the pages.** You can **use the opposite side of the page** for writing your answers.
- You can add additional pages if required, but you **must write your ID, Section and Name at the top of each additional page.**
- Please note, **Collaboration $\neq$ copying.** You are allowed to discuss the questions and clear confusions you might have, but you have to write your solutions independently and distinctively.
- **Scan** your answers, make a single PDF file by renaming "A3_StudentID_Fullname_CSE460.pdf" and **upload** it in the submission link.
- **Any findings of plagiarism will** result in a **Zero mark**. It may also hamper your future grading!
- **You must submit the online copy by the deadline.**

Question-1

2.
(CO1)

Consider the following truth tables for two logic gates

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

Gate 1

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

Gate 2

(a)	Identify the logic gates and draw their CMOS circuit representation.	2
(b)	Determine the individual transistor widths k_{nMOS} and k_{pMOS} for both of the gates to achieve the effective rise and fall resistances equal to that of a unit inverter, R , in the worst case.	2
(c)	Draw the simplified RC circuit only for Gate 1 and determine the input capacitance of Gate 2 in terms of C	4+1
(d)	Consider Gate 1 is driving 12 identical Gate 2 circuits (only one of the inputs of each). Now derive the expressions for t_{pdf} and t_{cdf} by sketching the corresponding RC networks from (c).	6

Question-2

2. (CO1) Brook is a chip designer who works specifically on the $\lambda = 1.5 \mu\text{m}$ process. However, he has messed up the layout design of a 3-input NOR gate (Figure 2(a)) because of which, the NOR gate has an equivalent rise resistance that is four times its equivalent fall resistance. Also, the NOR gate's equivalent rise resistance is twice the rise resistance of a unit inverter for that design process. But at least he made sure to keep the source-drain terminals shared where possible to reduce capacitance. The gate capacitance and parasitic (diffusion) capacitance of are $0.5 \text{ fF}/\mu\text{m}$.

This NOR gate is loaded with a Unit inverter and the output node Y has a voltage waveform as shown in Fig. 2(c).

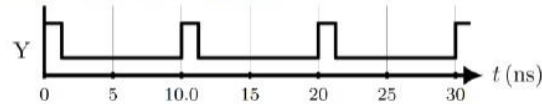


Figure 2(c): Voltage waveform at node Y

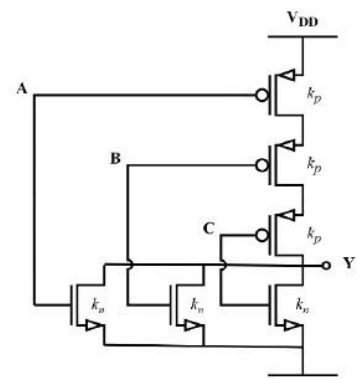


Figure 2(a): CMOS implementation of a 3 - input NOR gate

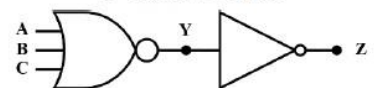


Figure 2(b): A 3-input NOR gate loaded with a Unit Inverter

- Find the value of width scaling factors - k_p and k_n of the NOR gate designed by Brook. Thereafter, calculate the actual width of the pMOS and nMOS. [2]
- Draw the RC equivalent model of the above 3-input NOR gate. Lump the capacitances together where possible. [5]
- Find the capacitance of the Y node (in fF units), when it is loaded with a unit inverter, as shown in Fig. 2(b). [3]
- Find the best-case falling edge contamination delay (t_{cdf}) of Y-node in terms of R and C of unit MOS. [2]
[Hint: You do not need Elmore modelling. All three PMOS are off. Three NMOS resistances are in parallel.]
- Find the switching power of the Y-node, if the system has a supply voltage of 5 V. [3]